Grade 12 Advanced Functions
Rate of Change Key

First Name: _____**KEY**_____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	Total
Points:	5	18	15	38
Score:				

Multiple Choice (10 marks)

1. (1 point) The average rate of change (AROC) of $f(x) = x^2 - 4x$ on $[1, 5]$ is:
   A. 1
   B. 2
   C. 4
   D. -2

2. (1 point) The instantaneous rate of change (IROC) of f at $x = a$ is equal to:
   A. $\dfrac{f(a+1) - f(a)}{1}$
   B. $\lim\limits_{h \to 0} \dfrac{f(a+h) - f(a)}{h}$ **(Newton's quotient limit)**
   C. $\dfrac{f(a) - f(0)}{a - 0}$
   D. $\dfrac{f(a+h) - f(a-h)}{2h}$ for any fixed h

3. (1 point) AROC over $[a, b]$ is the slope of the:
   A. Tangent line at $x = a$
   B. Tangent line at $x = b$
   C. Secant line through $(a, f(a))$ **and** $(b, f(b))$
   D. Horizontal line $y = f\left(\dfrac{a+b}{2}\right)$

4. (1 point) Evaluate the limit: $\lim\limits_{x \to 3} \dfrac{x^2 - 9}{x - 3}$.
   A. 3
   B. 6
   C. 9
   D. Does not exist

5. (1 point) For $f(x) = \sqrt{x+1}$, the IROC at $x = 3$ equals (using the limit of the difference quotient):
   A. $\dfrac{1}{4}$
   B. $\dfrac{1}{8}$
   C. $\dfrac{1}{2}$
   D. 1

Long Answer (33 marks total)

6. AROC and IROC via Newton's Quotient (10 marks) Let $f(x) = x^2 - 4x + 1$.
   (a) (4 points) Compute the average rate of change of f on $[1, 1+h]$ in terms of h (simplify fully).
   (b) (4 points) Use your result to find the instantaneous rate of change at $x = 1$ by taking the limit as $h \to 0$ (show every algebraic step).
   (c) (2 points) Interpret your numerical result from part (b) if f measures position (meters) and x is time (seconds).

Solution: (a) AROC on $[1, 1+h]$:

$$\textbf{AROC} = \frac{f(1+h) - f(1)}{(1+h) - 1} = \frac{f(1+h) - f(1)}{h}.$$

Compute:

$$f(1+h) = (1+h)^2 - 4(1+h) + 1 = 1 + 2h + h^2 - 4 - 4h + 1 = h^2 - 2h - 2, \quad f(1) = 1 - 4 + 1 = -2.$$

Thus

$$\frac{f(1+h) - f(1)}{h} = \frac{(h^2 - 2h - 2) - (-2)}{h} = \frac{h^2 - 2h}{h} = h - 2.$$

(b) IROC at $x = 1$ (limit of Newton's quotient):

$$\textbf{IROC} = \lim_{h \to 0} (h - 2) = -2.$$

(c) Interpretation: Instantaneous velocity at $t = 1$ s is -2 m/s (moving in the negative direction).

7. AROC and IROC from a Table (8 marks) A particle's position (meters) at times t (seconds) is:

t (s)	0	1	2	3	4	5
$s(t)$ (m)	0.0	1.1	3.8	7.7	12.5	18.0

   (a) (3 points) Compute the average rate of change on $[1, 4]$ with proper units.
   (b) (3 points) Estimate the instantaneous rate of change at $t = 3$ using a symmetric secant around $t = 3$.
   (c) (2 points) Briefly interpret both answers (speeding up, slowing down, or roughly constant?).

Solution: (a)

$$\textbf{AROC} = \frac{s(4) - s(1)}{4 - 1} = \frac{12.5 - 1.1}{3} = \frac{11.4}{3} = 3.8 \text{ m/s}.$$

(b) Symmetric about $t = 3$ (use $t = 2$ and $t = 4$):

$$\textbf{IROC} \approx \frac{s(4) - s(2)}{4 - 2} = \frac{12.5 - 3.8}{2} = \frac{8.7}{2} = 4.35 \text{ m/s}.$$

(c) IROC near 3 s is larger than AROC on $[1, 4]$, suggesting speed is increasing (speeding up) around $t = 3$.

8. Limits and IROC by Algebraic Simplification (9 marks) Consider $f(x) = \sqrt{x+1}$.
   (a) (5 points) Compute the instantaneous rate of change at $x = 3$ using the limit of the difference quotient. Do not use rules; rationalize carefully.
   (b) (2 points) Write the AROC over $[3, 3+h]$ in simplest form (in terms of h), showing the algebra.
   (c) (2 points) Evaluate your AROC formula for $h = 0.1$ and $h = -0.1$ and compare to your IROC value.

Solution: (a)

$$\textbf{IROC} = \lim_{h\to 0} \frac{\sqrt{4+h}-2}{h} = \lim_{h\to 0} \frac{\sqrt{4+h}-2}{h} \cdot \frac{\sqrt{4+h}+2}{\sqrt{4+h}+2} = \lim_{h\to 0} \frac{(4+h)-4}{h(\sqrt{4+h}+2)}.$$

$$= \lim_{h\to 0} \frac{h}{h(\sqrt{4+h}+2)} = \lim_{h\to 0} \frac{1}{\sqrt{4+h}+2} = \frac{1}{2+2} = \frac{1}{4}.$$

(b)

$$\textbf{AROC} = \frac{\sqrt{4+h}-2}{h} = \frac{1}{\sqrt{4+h}+2}.$$

(c) For $h = 0.1$: $\frac{1}{\sqrt{4.1}+2} \approx \frac{1}{4.0249} \approx 0.248$. **For $h = -0.1$:** $\frac{1}{\sqrt{3.9}+2} \approx \frac{1}{3.9748} \approx 0.252$. **Both are close to 0.25, matching IROC $= \frac{1}{4}$.**

9. Population of a Bacterial Colony using IROC (6 marks) A bacterial colony's population (cells) is modeled by $P(t) = t^2 + 6t + 200$, where t is time in hours.
   (a) (3 points) Compute the average rate of change of P on $[4, 7]$. Include units and interpretation.
   (b) (3 points) Find the instantaneous rate of change at $t = 5$ using the limit of the Newton's quotient (no rules). Interpret in context.

Solution: (a)

$$P(7) = 49 + 42 + 200 = 291, \quad P(4) = 16 + 24 + 200 = 240.$$

$$\textbf{AROC} = \frac{P(7) - P(4)}{7-4} = \frac{291-240}{3} = \frac{51}{3} = 17 \textbf{ cells/hour}.$$

Interpretation: On average, population increased by 17 cells per hour from 4 h to 7 h.

(b)

$$\textbf{IROC at } t = 5 = \lim_{h \to 0} \frac{P(5+h) - P(5)}{h}.$$

Compute each term:

$$P(5+h) = (5+h)^2 + 6(5+h) + 200 = 25 + 10h + h^2 + 30 + 6h + 200 = h^2 + 16h + 255,$$

$$P(5) = 25 + 30 + 200 = 255.$$

Difference quotient:

$$\frac{P(5+h) - P(5)}{h} = \frac{(h^2 + 16h + 255) - 255}{h} = \frac{h^2 + 16h}{h} = h + 16.$$

Limit as $h \to 0$:

$$\textbf{IROC} = 16 \textbf{ cells/hour}.$$

Interpretation: At $t = 5$ h, the instantaneous growth rate is 16 cells per hour (instantaneous increase at that moment).